AC Circuits & Power

$\overline{V}, \overline{I}$: RMS phasors with bars above. V, I: RMS magnitudes. V_m, I_m: peaks. Cosine-reference phasors; sine convention labeled where used.

Phasors & essential math

L2–3

$$\omega = 2\pi f = \frac{2\pi}{T}, \qquad 180^\circ = \pi \text{ rad}$$

$$\sin x = \cos(x - 90^\circ) \qquad \cos x = \sin(x + 90^\circ)$$
$$-\sin x = \sin(x \pm 180^\circ) \qquad -\cos x = \cos(x \pm 180^\circ)$$
$$\sin(-x) = -\sin x \qquad \cos(-x) = \cos x$$

$$z = a + jb = re^{j\phi} = r\angle\phi, \qquad j^2 = -1$$

$$r = \sqrt{a^2 + b^2}, \quad \tan\phi = \frac{b}{a}$$

$$a = r\cos\phi, \quad b = r\sin\phi, \quad z^* = a - jb$$

$$z_1 z_2 = r_1 r_2 \angle(\phi_1 + \phi_2)$$

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}\angle(\phi_1 - \phi_2)$$

Add/subtract in rectangular form; choose the correct angle quadrant. At the same frequency, larger phase leads.

$$V = \frac{V_m}{\sqrt{2}}, \quad I = \frac{I_m}{\sqrt{2}}, \quad \overline{V} = V\angle\phi$$

$$v(t) = \sqrt{2}V\cos(\omega t + \phi)$$

Impedance & circuit solution

L3–4

Element	Time domain	Z	$Y = 1/Z$
R	$v = Ri$	R	$1/R$
L	$v = L\,di/dt$	$j\omega L$	$-j/(\omega L)$
C	$i = C\,dv/dt$	$-j/(\omega C)$	$j\omega C$

$$\overline{V} = Z\overline{I}, \qquad Z = R + jX, \qquad Y = G + jB = \frac{1}{Z}$$

$$X_L = \omega L, \qquad X_C = \frac{1}{\omega C}, \qquad Z_{RLC} = R + j(X_L - X_C)$$

R: in phase. L: current lags voltage by 90°. C: current leads voltage by 90°. X_C is a positive magnitude.

$$\sum \overline{V}_k = 0 \quad \text{(KVL)}, \qquad \sum \overline{I}_k = 0 \quad \text{(KCL)}$$

$$Z_{\text{series}} = \sum_k Z_k, \qquad Y_{\parallel} = \sum_k Y_k$$

$$Z_1 \parallel Z_2 = \frac{Z_1 Z_2}{Z_1 + Z_2}$$

$$\overline{V}_2 = \frac{Z_2}{Z_1 + Z_2}\overline{V}_{\text{in}}$$

$$\overline{I}_1 = \frac{Z_2}{Z_1 + Z_2}\overline{I}, \qquad \overline{I}_2 = \frac{Z_1}{Z_1 + Z_2}\overline{I}$$

$$Z_{\text{Th}} = Z_{\text{N}}, \qquad \overline{V}_{\text{Th}} = \overline{I}_{\text{N}} Z_{\text{Th}}$$

Equivalent impedance: short independent voltage sources; open independent current sources.

Series RL sinusoidal response

L3

For $v_s = V_m\cos(\omega t + \phi)$, with $i(0) = 0$:

$$A = \frac{V_m}{\sqrt{R^2 + (\omega L)^2}}, \qquad \theta = \tan^{-1}\frac{\omega L}{R}$$

$$i(t) = \underbrace{A\cos(\omega t + \phi - \theta)}_{\text{steady state}} - \underbrace{A\cos(\phi - \theta)e^{-Rt/L}}_{\text{transient}}$$

Power for a series equivalent $R + jX$

V: total RMS voltage across $R + jX$; use signed X.

$$P = I^2 R = \frac{V^2 R}{R^2 + X^2}, \qquad Q = I^2 X = \frac{V^2 X}{R^2 + X^2}$$

AC power & power factor

L4–5

$$\theta = \theta_v - \theta_I, \qquad p(t) = v(t)i(t)$$

Sine convention: $v = V_m\sin\omega t$, $i = I_m\sin(\omega t - \theta)$.

$$p(t) = VI[\cos\theta - \cos(2\omega t - \theta)]$$

$$P = VI\cos\theta \quad \text{[W]}$$
$$Q = VI\sin\theta \quad \text{[var]}$$
$$\overline{S} = \overline{V}\overline{I}^* = P + jQ \quad \text{[VA]}$$
$$S = |\overline{S}| = \sqrt{P^2 + Q^2} = VI$$

$$\text{PF} = \cos(\theta_v - \theta_I) = \cos\theta = \frac{P}{S}$$

$$I = \frac{\sqrt{P^2 + Q^2}}{V}$$

$$\overline{I} = \left(\frac{\overline{S}}{\overline{V}}\right)^*$$

If the given phasors use peak magnitudes: $\overline{S} = \frac{1}{2}\overline{V}_{\text{peak}}\overline{I}^*_{\text{peak}}$.

Load	Q, θ	Current / pf
Inductive	> 0	Lagging
Capacitive	< 0	Leading
Resistive	$= 0$	In phase / unity

Component-voltage formulas: use the RMS voltage across each individual element, not the total voltage across a series combination.

$$P_R = I_R^2 R = \frac{V_R^2}{R}$$

$$Q_L = I_L^2 X_L = \frac{V_L^2}{X_L}, \qquad Q_C = -I_C^2 X_C = -\omega C V_C^2$$

Line voltage & power balance

L4–5

$$\overline{V}_s = \overline{V}_{\text{load}} + \overline{I}Z_{\text{line}}$$

$$P_s = P_{\text{load}} + P_{\text{loss}}, \quad P_{\text{loss}} = I^2 R_{\text{line}}$$

$$Q_s = Q_{\text{load}} + Q_{\text{line}}, \quad Q_{\text{line}} = I^2 X_{\text{line}}$$

Capacitor for a target power factor

L5

Shunt capacitor at the load. For an initially lagging load and a lagging/unity target, let PF_1 be initial and $\text{PF}_2 \geq \text{PF}_1$ target.

$$C = \frac{P\left[\tan(\cos^{-1}\text{PF}_1) - \tan(\cos^{-1}\text{PF}_2)\right]}{\omega V^2}$$

P: load real power [W]; V: load-bus RMS voltage [V]; $\omega = 2\pi f$; C in farads. Use P, V at the same operating point.

If the original load is given as $R + jX_L$:

$$C = \frac{X_L - R\tan(\cos^{-1}\text{PF}_2)}{\omega(R^2 + X_L^2)}$$

For unity pf, set $\text{PF}_2 = 1$. For a leading target, use $-\cos^{-1}\text{PF}_2$ as its target angle.

Original lecture example ($f = \text{PF}_2$, lagging):

$$Y_{\text{new}} = \frac{R}{R^2 + X_L^2} - j\left(\frac{X_L}{R^2 + X_L^2} - \frac{1}{X_C}\right) = a - jb$$

$$\frac{b}{a} = \tan(\cos^{-1} f), \qquad X_C = \frac{1}{\omega C}$$

Converters & Electricity Demand

Ideal devices. V_m is AC source peak; K duty ratio, τ switching period, V_t converter output. Angles α in formulas below are radians.

Periodic waveforms L4,6

$$V_{\text{av}} = \frac{1}{T}\int_0^T v(t)\,dt, \qquad V_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\,dt}, \qquad v(t) = V_{\text{av}} + \text{ripple}$$

$$P_R = \frac{V_{\text{rms}}^2}{R} = I_{\text{rms}}^2 R, \qquad I_{\text{av}} = \frac{V_{\text{av}}}{R}$$

Rectifiers: final average, RMS, and power L6

$v_s = V_m \sin \omega t$; resistive load, no filter; $0 \le \alpha \le \pi$.

	Half-wave	Full-wave
V_{av}	$\frac{V_m}{2\pi}(1+\cos\alpha)$	$\frac{V_m}{\pi}(1+\cos\alpha)$
V_{rms}	$\frac{V_m}{2}\sqrt{1-\frac{\alpha}{\pi}+\frac{\sin 2\alpha}{2\pi}}$	$\frac{V_m}{\sqrt{2}}\sqrt{1-\frac{\alpha}{\pi}+\frac{\sin 2\alpha}{2\pi}}$
P_R	$\frac{V_m^2}{8\pi R}[2(\pi-\alpha)+\sin 2\alpha]$	$\frac{V_m^2}{4\pi R}[2(\pi-\alpha)+\sin 2\alpha]$
Diode: $\alpha = 0$	$V_{\text{av}} = V_m/\pi, \quad V_{\text{rms}} = V_m/2$	$V_{\text{av}} = 2V_m/\pi, \quad V_{\text{rms}} = V_m/\sqrt{2}$

DC/DC converters L5–7

$$K = \frac{t_{\text{on}}}{\tau}, \qquad \tau = t_{\text{on}} + t_{\text{off}} = \frac{1}{f_s}, \qquad P_{\text{av}} = KP_{\text{on}}$$

Buck / chopper: resistive, unfiltered load

$$V_{t,\text{av}} = KV_s, \qquad V_{t,\text{rms}} = V_s\sqrt{K}$$

Boost: filtered output, continuous inductor current

$$V_t = V_s\left(1+\frac{t_{\text{on}}}{t_{\text{off}}}\right)$$

$$\Delta i = \frac{V_s t_{\text{on}}}{L} \qquad \Longleftrightarrow \qquad L = \frac{V_s t_{\text{on}}}{\Delta i}$$

Inverting buck-boost: filtered output

$$V_t = -V_s\frac{t_{\text{on}}}{t_{\text{off}}}$$

Boost/buck-boost: periodic steady state, continuous inductor current, small output ripple. Δi is peak-to-peak current ripple.

$$P_{\text{in}} = P_{\text{out}}, \qquad V_s I_s = |V_t|\,|I_t|$$

Current magnitudes in the power balance; $|I_t| = |V_t|/R$ for the resistive load.

Switches & motor-drive relations L5–7

Ideal switch: ON $v_{\text{sw}} = 0$; OFF $i_{\text{sw}} = 0$. SCR turns on with forward bias + gate pulse, and turns off when $i_A < I_H$.

$$n \propto f, \qquad P_{\text{pump/fan}} \propto n^3$$

Rectifier with a shunt capacitor L7

SCR ON: capacitor follows the source

$$v_t = v_C = v_s, \qquad v_{\text{SCR}} = 0$$

$$i_A = \frac{V_m}{R}[\sin\omega t + \omega RC\cos\omega t]$$

SCR OFF: capacitor supplies the resistor

$$i_A = 0, \qquad v_C(t) = v_C(0)e^{-t/(RC)}$$

Ideal turn-off occurs when $i_A = 0$. With a series inductor instead, current cannot jump and conduction continues while current is positive.

Basic AC/AC controller L7

$$V_{\text{av}} = 0, \qquad V_{\text{rms}} = \frac{V_m}{\sqrt{2}}\sqrt{1-\frac{\alpha}{\pi}+\frac{\sin 2\alpha}{2\pi}}$$

Resistive load, no filter. Output frequency = input frequency; use $P_R = V_{\text{rms}}^2/R$.

H-bridge square-wave inverter L7

$$v_{AB} = \pm V_s, \qquad V_{\text{av}} = 0, \qquad V_{\text{rms}} = V_s$$

Equal positive/negative half-cycles.

Demand, energy & billing L8

$$P(t) = \sum_i P_i(t), \qquad Q(t) = \sum_i Q_i(t)$$

$$E = \int_t^{t+T} P(u)\,du = \sum_k P_k \Delta t_k, \qquad P_{\text{av}} = \frac{E}{T}$$

$$\text{peak/average} = \frac{P_{\text{peak}}}{P_{\text{av}}}, \qquad \text{CF}_{\text{lecture}} = \frac{P_{\text{av}}}{P_{\text{peak}}}$$

$$A = c_E E + FC \qquad \text{(energy + fixed charge)}$$

$$A = c_E E + c_D P_{\text{peak}} + FC \qquad \text{(with demand charge)}$$

E [kWh], Δt [h], c_E [\$/kWh], c_D [\$/kW per billing period]. Add a pf penalty only if specified.

Load-duration curve L8

Sort load samples high to low, retaining their durations. Curve gives hours for which load exceeds a level; area is total energy. Aggregation lowers the peak relative to the sum of individual peaks.